PHARMACY DEPARTMENT · PHARMACEUTICAL PRODUCT REGISTRATION

دائرة الصيدلة · تسجيل المستحضرات الصيدلانية

INTERNAL · FOR DEPARTMENTAL REVIEW | PPR-CMP-001 · VERSION 0.2 · 21 OCTOBER 2026

Comparative Assessment Report

تقرير التقييم المقارن

PPR and EXTEDO — a side-by-side comparison, updated

The PPR medicine approval system compared with the EXTEDO eCTD validation and review products used by regulatory authorities, revised after three weeks of development that closed most of the gaps the first comparison found — now with a side-by-side look at the two systems' screens.

تقرير مقارنة محدَّث: نظام تسجيل المستحضرات الصيدلانية (PPR) مقابل حلول EXTEDO للتحقق من ملفات eCTD ومراجعتها، مع مقارنة بصرية مباشرة بين واجهتي النظامين.

Report status	Supersedes	Gaps closed since v0.1
Draft for review	**PPR-CMP-001 v0.1 (18 Oct 2026)**	**7 of 12**

Document reference	PPR-CMP-001
Version / status	0.2 · Draft for review
Date of issue	21 October 2026
Prepared for	Head of Department, Pharmacy
Prepared by	PPR project team
Related documents	PPR-POC-001 Proof of Concept Report v0.2 (21 Oct 2026) · PPR-VAL-URS, PPR-VAL-RA, PPR-VAL-QP (docs/validation/)
Classification	Internal

REVISION HISTORY

0.1	18 Oct 2026 — First issue of the comparison report
0.2	21 Oct 2026 — Updated validation, workflow and platform sections for gaps closed since v0.1; added the visual (screenshot) comparison section; refreshed the gap analysis and recommendation

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10 Options and recommendation

A Sources

What changed in this version. Since the first comparison (18 October 2026), PPR adopted the official ICH eCTD 3.2 DTD and implemented the GCC Module 1 v1.5 validation criteria as real rules, added background (queued) validation, a printable short/long PDF validation report, a cumulative "current dossier" lifecycle view, full-text search across every dossier a reader may open, a key-protected read-only API, optional staff single sign-on, and a full set of system-validation documents (URS, risk assessment, IQ/OQ/PQ protocol, automated traceability). Section 9 tracks each of the twelve original gaps against this progress. The one item still outstanding that matters most for go-live is executing the validation benchmark against a real reference validator — the procedure is written, but PPR has no EURSvalidator licence to run it against.

1 Executive summary

This report compares **PPR**, the department's in-house medicine approval system, with the **EXTEDO** products that regulatory authorities use to receive, validate and review eCTD dossiers: EURS (desktop review system), EURSnext (its web companion, sold only to authorities) and EURSvalidator (the validation engine). It updates the first comparison (PPR-CMP-001 v0.1, 18 October 2026) after three weeks in which the project team closed most of the high-priority gaps that report identified, and adds a new side-by-side look at what each system's screens actually show (section 7).

The two remain different kinds of system, so they still overlap only in part:

EXTEDO EURS is a mature, specialised validation and dossier-review tool. By EXTEDO's account, more than 35 authorities use it, including the European Medicines Agency (EMA). EXTEDO also names Oman, Saudi Arabia, Bahrain, Qatar and the UAE among the agencies that use or recommend EURS or EURSvalidator. Its strengths are official, regularly updated validation criteria for each region (GCC included), deep document-review tools (annotation, full-text search, lifecycle navigation) and enterprise scale.

PPR is an end-to-end registration workflow system: a company portal, automated eCTD validation, assignment of specialist reviewers per CTD module, information requests, the HOD's decision, and a bilingual approval certificate. Its interface is in Arabic and English, it has an AI dossier assistant, and the department owns it.

7	2	3
GAPS CLOSED	IN PROGRESS / PARTIAL	STILL OPEN

Key finding. The balance has shifted since v0.1. PPR's validator now runs against the **official ICH eCTD 3.2 DTD** (checksum-verified) and implements the **GCC Module 1 v1.5** criteria as explicit rules — the two items the first report flagged as most important before go-live are done in code. PPR also gained background (queued) validation, a printable short/long PDF report, a cumulative dossier view, department-wide search, an integration API and optional staff single sign-on. EXTEDO remains ahead on reach and proof: validation sets kept current across every region by the vendor, true in-PDF annotation, NeeS and eCTD 4.0 support, and years of production use at 35+ authorities including the EMA. PPR remains the only one of the two that runs the whole registration case — company portal, validation, specialist review, decision and certificate — in one system.

What is not yet done. PPR's validation criteria have not been independently checked against a real reference validator: the benchmark procedure is written (`docs/validation_benchmark.md`) and the tooling exists, but the department has no EURSvalidator evaluation licence to run it against, so this has not been executed. The new system-validation documentation (URS, risk assessment, IQ/OQ/PQ protocol) is drafted and every requirement is traced to an automated test, but qualification execution and the HOD/QA/IT sign-off have not happened. The GCC regional DTD PPR validates against is still transcribed from the published specification text, because SFDA does not distribute the file itself.

Recommendation (section 10, unchanged in substance). Keep PPR as the registration and workflow system. The two most important remaining steps are to run the validation benchmark against EURSvalidator (or another reference validator) and to execute and sign off the qualification protocol, before go-live. If budget allows, an integration in which PPR manages the case and calls an EXTEDO validator or review tool remains a viable fallback if the benchmark turns up material gaps.

2 Scope, method and sources

What was compared

PPR: the code in the project repository as of 21 October 2026, including uncommitted work in progress at the time of writing. This update reviewed the README, data models, validation rules (`apps/validation/`), workflow, tests, and the new validation documentation set in `docs/validation/` (URS, risk assessment, IQ/OQ/PQ protocol, traceability matrix) and `docs/validation_benchmark.md`.

EXTEDO: the products offered to regulatory authorities (EURS, EURSnext, EURSvalidator), plus eCTDmanager where the applicant side is relevant. The wider EXTEDOpulse RIM suite (registration, safety, quality and document hubs) is mentioned only where it fills a gap. This update re-checked EXTEDO's current public pages; their structure has changed since v0.1 under EXTEDO's "a Cormeo brand" refresh, but the products and their described capabilities have not.

Method and limitations

- EXTEDO was assessed from public material only: product information sheets, EXTEDO web pages and announcements, and the EMA eSubmission page. No EXTEDO software was installed or tested, and no pricing was available. This has not changed since v0.1.
- Most EXTEDO claims come from the vendor. Customer counts differ between sources (25, 30 and 35+ authorities). They are quoted as EXTEDO's own claims.
- "Not stated" in this report means that the feature was not described in the material reviewed. It does not mean that EXTEDO lacks it. These points should be confirmed with the vendor.
- PPR is a proof of concept. Statements about PPR describe what is in the code and its tests, not what has been proven in a production deployment. Where a PPR claim is backed by an automated test or a specific file, this report names it so a reader can verify it directly.
- **New in this version:** section 7 places real PPR screenshots (captured 21 October 2026 from the running application with demonstration data) next to EXTEDO material. EXTEDO's current marketing pages carry no embedded screenshots of the live product; where a real vendor screenshot could not be obtained, the EXTEDO side is shown as a short, sourced description from vendor material instead of an image — this is marked clearly on each pair and is not a simulated or reconstructed screenshot.

3 The two systems at a glance

PPR · Pharmaceutical Product Registration

Type	Web-based registration workflow system with built-in eCTD validation
Built by	PPR project team for the Pharmacy Department (in-house)
Users	Company applicants, specialist reviewers, the lead reviewer, the HOD and an administrator
Technology	Django, Tailwind CSS, SQLite locally and MySQL on PythonAnywhere
Languages	English and Arabic (right-to-left)
Validation	31 rules in 3 layers (XML schema, business rules incl. GCC Module 1 v1.5, file compliance), against the official ICH eCTD 3.2 DTD
Status	Proof of concept with an automated test suite (23 traced requirements); qualification protocol drafted but not yet executed; not yet in production

EXTEDO · EURS, EURSnext, EURSvalidator

Type	Commercial eCTD/NeeS validation and review software for authorities
Vendor	EXTEDO (Germany), a Regulatory Information Management (RIM) vendor, now "a Cormeo brand"
Users	Agency validators and assessors, including external assessors (EURSnext)
Technology	Windows desktop and server services; EURSnext runs in the browser
Languages	English; Arabic is not mentioned in the material reviewed
Validation	Official validation sets per region, including all GCC member states, sold as subscription packages, selected automatically
Status	Mature product; selected for EMA use and offered to EU national authorities

Where each system fits in the regulatory process

Process step	PPR	PPR	EXTEDO	EXTEDO (authority-side products)
Applicant builds the dossier	—	Not covered. Companies build the eCTD with their own tools.	✓	eCTDmanager (applicant product) builds, validates and publishes dossiers.
Applicant submits	✓	Company portal: create the application, upload the sequence ZIP, submit.	~	Via gateway: receipt through CESP connect, EMA Common Repository connector, server monitoring.
Technical validation	✓	Automatic on submission or queued in the background; 31 rules against the official DTD and GCC M1 v1.5.	✓	EURSvalidator with official regional validation sets, auto-selected.
Scientific review	✓	Assessment per CTD module by a named specialist, recorded in the	✓	Viewing, search, annotation and export of annotations for the

		system.		assessment report.
Case management (assignment, questions, deadlines)	✓	Automatic team assignment by workload, information requests, status tracking, notifications.	?	Not stated: filtering by regulatory activity; case workflow through the API or other EXTEDO hubs.
Decision and certificate	✓	HOD approves or rejects; bilingual certificate with validity period.	—	Not part of EURS.

4 Capability coverage matrix

Full available and substantial · Partial available but limited · None not available · Not stated not described in the public material · changes since v0.1 are marked ▲

Capability	PPR	EXTEDO	Comment
VALIDATION			
eCTD v3.2.2 validation	Full ▲	Full	PPR now validates against the official ICH DTD, checksum-pinned (was: reference copy)
GCC Module 1 validation	Full ▲	Full	PPR implements GCC M1 v1.5 as rules (was: existence/well-formedness only). Regional DTD is still transcribed from the spec, not SFDA's own file
NeeS and other formats	None	Full	PPR accepts eCTD only — unchanged
eCTD 4.0 readiness	None ▲	Partial	PPR now recognises an eCTD 4.0 package and refuses it with a clear message (was: no handling); still no validation support. EXTEDO says it is "prepared for" eCTD 4.0
PDF compliance checks	Full	Full	PPR: version, security, JavaScript, fonts — unchanged
Validation report export	Full ▲	Full	PPR now adds a printable short/long PDF report alongside on-screen and CSV (was: on-screen/CSV only). EXTEDO additionally auto-saves and emails the PDF itself
REVIEW			
CTD tree and dossier viewer	Full	Full	Unchanged
Lifecycle (cumulative) view	Full ▲	Full	PPR's "Current dossier" tab now applies replace/append/delete across all accepted sequences (was: sequence-by-sequence only)
PDF annotation	Partial ▲	Full	PPR adds page-linked review notes, quoting selected text, inserted into the module assessment (was: none). Still not in-PDF highlighting/markup the way EXTEDO describes it
Full-text search	Full ▲	Full	PPR now searches every dossier a reader may open, not just the one open in the assistant (was: single-dossier only)
AI dossier assistant	Full	Not stated	PPR: cited answers, offline mode without a key; a smoke-test command exists but a live-model run has not been recorded
WORKFLOW AND ADMINISTRATION			
Applicant portal	Full	None	EXTEDO receives dossiers through gateways — unchanged
Specialist assignment per module	Full	Not stated	PPR: automatic by workload is now the default, with manual HOD assignment as a fallback
Information requests	Full	Not stated	Unchanged

Decision and certificate	Full	None	Unchanged
KPIs and dashboards	Full ▲	Not stated	PPR: HOD dashboard with status breakdown, reviewer workload and average days to decision — a complete departmental KPI view, not an enterprise BI suite
Audit trail	Full	Full	Unchanged
Arabic interface	Full	Not stated	Unchanged — see section 7 for a real screenshot
PLATFORM			
Integration API / SSO	Partial ▲	Full	PPR now has a key-protected read-only JSON API and optional staff OpenID Connect SSO (was: none). No write API, Active Directory binding or document-management connectors yet
Scale and parallel processing	Partial ▲	Full	A background validation queue now decouples large uploads from the web request (was: fully synchronous); still a single web application, not distributed services
Formal system validation	Partial ▲	Full	URS, risk assessment, IQ/OQ/PQ protocol and an automated traceability matrix are now drafted and every requirement is tested (was: none); execution and sign-off are still outstanding

5 Detailed comparison

Area	PPR	EXTEDO
Primary purpose	Manage a marketing authorisation application from submission to certificate, with validation built in.	Validate, accept, import, review and maintain eCTD and NeeS submissions inside an authority.
Submission intake	Companies upload one ZIP per sequence (0000, 0001, ...) through the portal. Safety checks on the ZIP: no zip-slip, symlinks, encryption or zip bombs.	Import from gateways: CESP connect (automatic import of CESP/FTP receipts), CR Connect (EMA Common Repository), and server monitoring that unzips and imports automatically.
Application types	New drug, generic, variation, renewal. The modules required for each type are enforced by rule BR-06.	Any standard submission format and regulatory activity; filtering by activity and by product status.
Validation engine	31 rules in 3 layers against the official ICH eCTD 3.2 DTD and the GCC Module 1 v1.5 criteria, each rule a small class so new ones are easy to add. Runs synchronously in development or in a background queue in production (`VALIDATION_MODE`); any error fails the sequence, warnings let it pass.	EURSvalidator applies official validation sets that EXTEDO keeps current in cooperation with authorities. The validation set and DTD are selected automatically. Batch validation of all pending submissions.
Validation results	Stored per run with findings; shown as a report with CSV export and a printable short/long PDF report, linked from the validation email.	Short PDF report (errors) and long PDF report (errors, warnings and details); reports are saved automatically and can be emailed.
Dossier navigation	CTD tree for each accepted sequence, plus a cumulative "Current dossier" view applying every replace/append/delete across sequences, with links to the documents (served through access-controlled views).	Navigation through the whole lifecycle of a product, envelope information, and table-of-contents elements.
Document review tools	PDFs open in the browser. Reviewers add page-linked notes, optionally quoting selected text, collected for the assessment. The assistant can search and read documents and cites `sequence/path p.N` .	Annotate, highlight and hyperlink PDFs without changing the originals; export annotations to support the assessment report; full-text search across documents, envelopes, TOC elements and annotations.
Review organisation	The HOD sets each reviewer's specialties (M1–M5); the review team is now assembled automatically by workload (least-loaded qualified reviewer per module, with a default lead), or by hand when a module has no qualified reviewer. Reviewers write assessments only for their own modules; the lead sends questions and the recommendation once every module is assessed.	Parallel work by many users. Not stated: per-module assignment, recommendation or decision workflow.
Communication with applicants	Information requests with due dates; the company answers with a validated response sequence. In-app and email notifications.	Not stated for EURS (handled outside the review tool).
Decision	The HOD approves (bilingual certificate with a validity period), rejects, or returns the application to the lead.	Not part of EURS.

AI assistance	Dossier assistant on every page: knows the current application and document, uses read-only tools limited to what the reader may see, and runs in offline mode (search passages) without an API key; a smoke-test command checks a configured model's citations.	Not stated in the material reviewed.
Security and access	Role-based access, documents served only to authorised users, audit trail in `StatusHistory`, plus optional OpenID Connect single sign-on for staff.	Extensive audit trail of user activity; Active Directory integration; optional storage in a document management system (Documentum, FileNet).
Integration	Key-protected read-only JSON API for other Ministry systems, plus optional SSO (was: Django admin only).	REST API, command-line automation engine, Web Broker for task queues, connectors for portals and document management systems.
Scale	Single web application on PythonAnywhere; large uploads and validation now run through a background worker queue, decoupled from the web request.	Scales beyond 400 users; background Windows services and load balancing for long imports and validations.
Deployment	Web; PythonAnywhere with MySQL. No client software.	EURS: desktop client with server services, installed on site. EURSnext: browser access, suited to external assessors.
Language	English and Arabic, with a full right-to-left layout.	English; Arabic not mentioned.
Cost and ownership	Owned by the department; hosting costs and staff time. Can be changed freely.	Commercial licences and validation-set subscriptions, priced per region (prices not public). Updates to standards are the vendor's job.
Validation documentation	URS (23 requirements), risk assessment (12 hazards) and an IQ/OQ/PQ qualification protocol are drafted, with every requirement traced to automated tests. Execution and HOD/QA/IT sign-off have not happened yet.	Not stated in the material reviewed; EXTEDO's flyers state the product is validated by the vendor and "installed and validated within days" at a customer site.
Maturity and references	Proof of concept with automated tests and a drafted validation pack; no production use yet.	Selected for EMA use and offered to EU national authorities; EXTEDO reports more than 35 authority customers, including GCC agencies.

6 eCTD technical validation in depth

Technical validation was the HOD's core requirement, so it is compared in more detail. Both systems follow the same principle as the FDA and EMA: an automated technical check must pass before scientific review starts.

PPR validation rules

Layer	Rules	What is checked
1. XML schema	SCH-01 to SCH-04, GCC-01 to GCC-02	`index.xml` exists, is well-formed and is valid against the official ICH eCTD 3.2 DTD; the regional Module 1 XML exists, is well-formed and its root is `gc:gc-backbone`.
2. Business rules	BR-01 to BR-10, GCC-03 to GCC-09	Sequence numbering; `index-md5.txt`; every leaf points to an existing file with a matching MD5; no unreferenced files; required modules for the application type; lifecycle operations resolve to earlier accepted sequences; ICH file naming and path length; the GCC envelope (receiving agency, submission type and unit, related sequences) and the required Module 1 documents for Oman, including Arabic and English leaflets.
3. File compliance	FC-00 to FC-08	Safe ZIP; accepted file formats (DOCX only in Module 1); PDFs not password-protected; PDF version 1.4–1.7; no JavaScript or launch actions; embedded fonts (warning); size limit; no empty files; searchable text rather than scanned images (warning).

How this compares with EURSvalidator

Where PPR is now on par

- **The official ICH eCTD 3.2 DTD**, checksum-verified against the published file (was a reference copy in v0.1)
- **The GCC Module 1 v1.5 criteria** implemented as rules: regional structure, envelope fields, submission type and unit, required Module 1 documents
- Checksums of the backbone and of every leaf
- Broken references and unreferenced files
- Lifecycle operations against earlier sequences
- PDF security, version, JavaScript and fonts
- File naming and path-length rules
- Clear separation of errors and warnings

Where EURSvalidator still goes further

- Validation sets kept current with every specification change, in cooperation with authorities, across **every** region EXTEDO supports — not only the GCC
- Automatic selection of the validation set and DTD across those regions
- NeeS, and actual preparation for eCTD 4.0 (PPR only detects and refuses an eCTD 4.0 package)
- PDF reports that auto-save and can be emailed directly from the tool
- The GCC regional DTD itself: EXTEDO's is the authority-distributed file; PPR's is still transcribed from the specification text, because SFDA does not publish the file
- Years of production use and, by EXTEDO's account, independent validation at 35+ authorities

Practical risk, updated. PPR's validation criteria are now aligned with the published GCC Module 1 v1.5 specification and the official ICH DTD — the design gap from v0.1 is closed. What remains is **confirmation**: this alignment has not been independently checked against EURSvalidator or another reference validator on the same dossiers. The benchmark procedure and tooling are ready (`docs/validation_benchmark.md`, `manage.py benchmark_validation`, automated regression tests in `apps/validation/benchmark.py`), but the department has no EURSvalidator evaluation licence to run it against, so PQ-02 in the qualification protocol is still open. GCC applicants typically validate their sequences with EXTEDO tools before submitting; until the benchmark runs, there remains a residual chance that a dossier which passed the applicant's check could fail in PPR, or the reverse. This is now the single most important step before go-live.

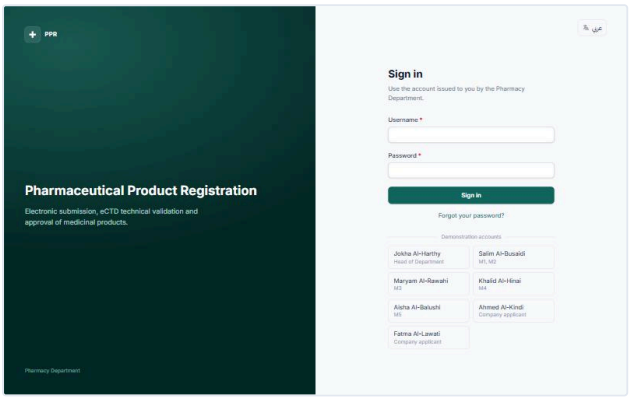
7 Visual comparison: PPR and EXTEDO side by side

PPR's screens below are real screenshots, captured on 21 October 2026 from the running application using demonstration data (fictional companies and products seeded for review — no real applicant information). EXTEDO's current marketing pages do not carry screenshots of the live product, so for most pairs the EXTEDO side is shown as a short, directly sourced description instead of a reconstructed or simulated image. Any EXTEDO screenshot that does appear is reproduced from EXTEDO's own published material and credited to it — it is not an independent test of the product (see section 2).

A · Sign-in and branding

PPR: live screenshot, 21 Oct 2026

PPR



Bilingual branding, one-click demonstration accounts for every role, language switch top-right.

EXTEDO

No independently obtained screenshot — EXTEDO software was not installed or tested, and the vendor's current marketing pages carry no embedded product screenshots.

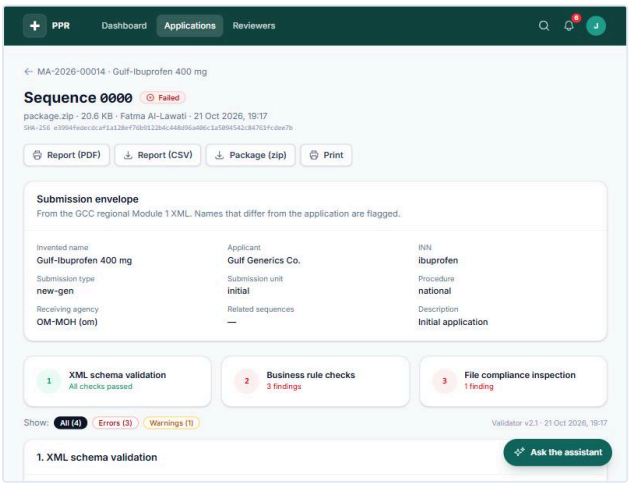
From vendor material:
EURS runs as a Windows desktop client with server services, installed on site; EURSnext gives browser access, built for external assessors. English interface — Arabic is not mentioned in the material reviewed.

Source: EXTEDO EURS Product Information (flyer); EXTEDO EURSnext product page

B · Technical validation report

PPR: MA-2026-00014, sequence 0000 — 3 errors, 1 warning

PPR



Three-layer summary, SHA-256 of the package, GCC envelope fields, and filterable findings by severity — this example fails on business-rule and file-compliance checks.

EXTEDO

No independently obtained screenshot.

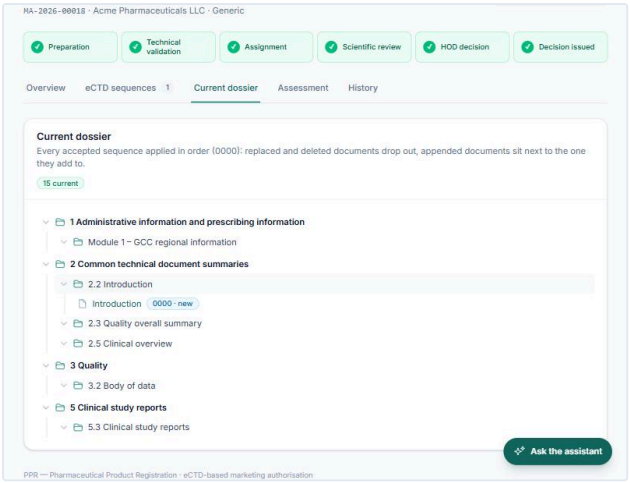
From vendor material:
EURSvalidator issues a short PDF report (errors) and a long PDF report (errors, warnings and full detail); both save automatically and can be emailed. The validation set and DTD are selected automatically for the detected region.

Source: EXTEDO EURSvalidator Product Information (flyer)

C · Dossier and CTD navigation

PPR: MA-2026-00018, current (cumulative) dossier

PPR



The CTD tree after every accepted sequence is applied in order; replaced and deleted documents drop out automatically.

EXTEDO

No independently obtained screenshot.

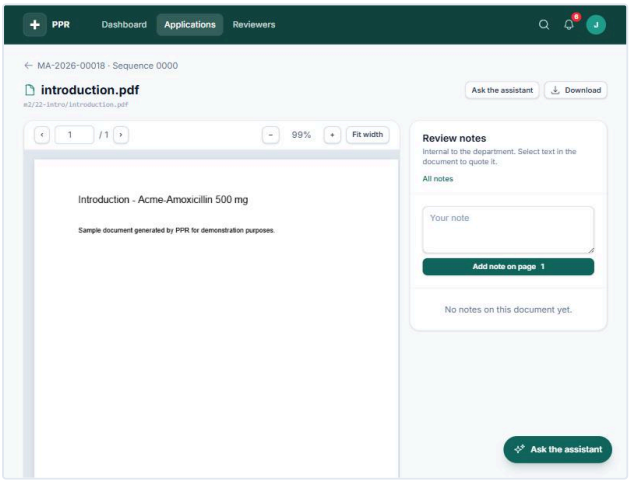
From vendor material:
EURS navigates the whole lifecycle of a product — envelope information, table-of-contents elements and every prior sequence — from the Windows desktop client or the EURSnext browser client.

Source: EXTEDO EURS Product Information (flyer)

D · Document review

PPR: in-app PDF.js viewer with review notes

PPR



PDFs open in the browser with page navigation and zoom; a reviewer's note links to a page and an optional text quote, without touching the file.

EXTEDO

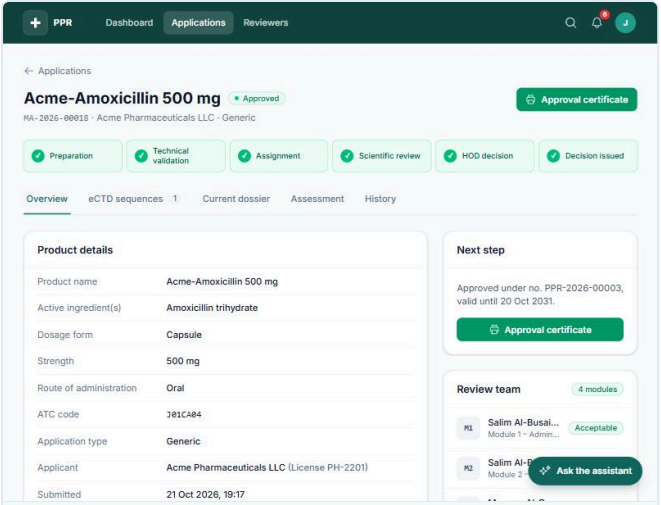
No independently obtained screenshot.

From vendor material:
reviewers annotate, highlight and hyperlink PDFs without changing the original files; full-text search spans documents, envelopes, TOC elements and annotations; annotations export to support the assessment report.

Source: EXTEDO EURS Product Information (flyer)

PPR capabilities with no EXTEDO equivalent in the material reviewed

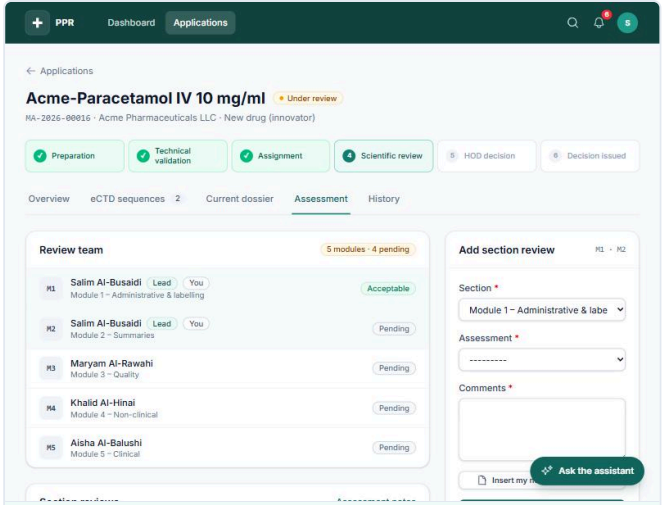
EURS and EURSnext validate and review submissions inside an authority; they do not run the registration case itself. The six screens below have no counterpart in the vendor material because EXTEDO's own flyers describe this ground as outside EURS (see sections 3–5).



The whole case in one place

A single timeline from preparation to decision issued, with the review team and next step always visible.

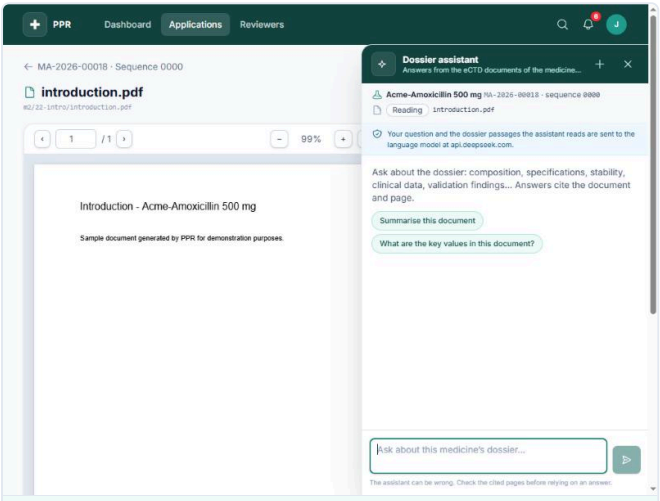
EXTEDO: "not part of EURS"



Per-module specialist assessment

Each CTD module goes to one named specialist; the lead reviewer records the team's progress and recommendation.

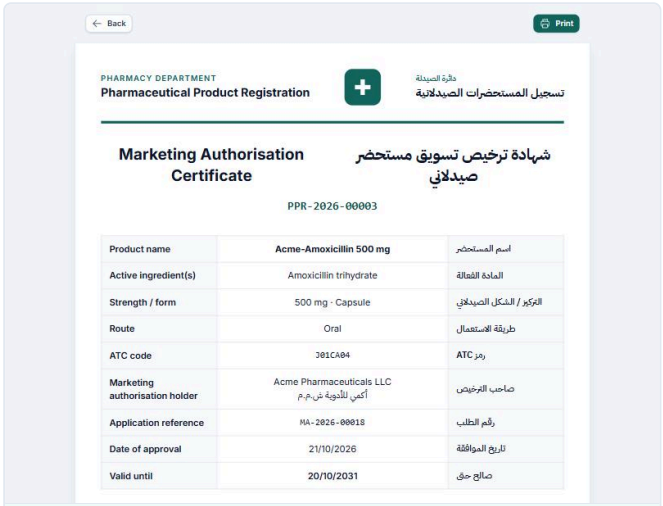
EXTEDO: "not stated" — no per-module workflow described



AI dossier assistant

Answers questions from the dossier the reader may see, citing the document and page; shows which host receives the text.

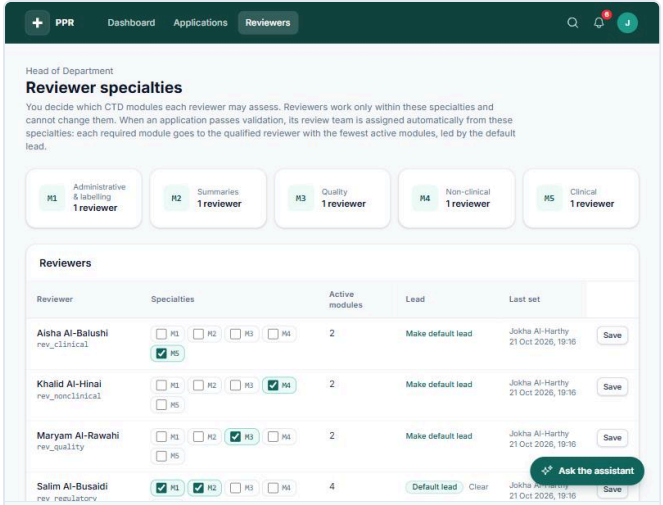
EXTEDO: "not stated" in the material reviewed



Bilingual approval certificate

Issued automatically on HOD approval, with a validity period, in English and Arabic.

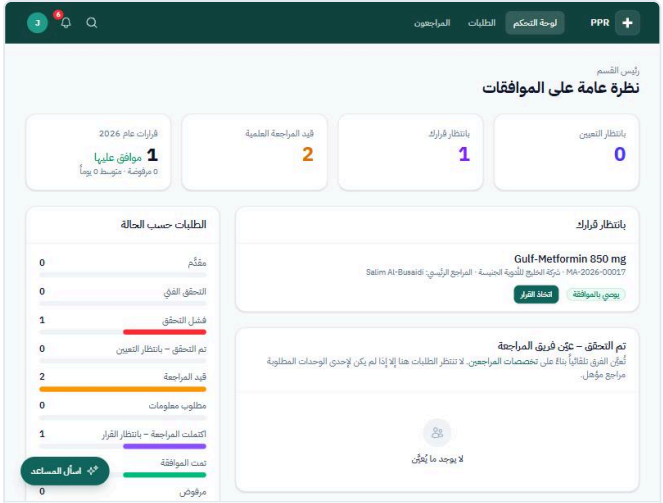
EXTEDO: "not part of EURS"



Reviewer specialties and workload

The HOD sets which CTD modules each reviewer may assess; teams are assembled automatically from this grid.

EXTEDO: "not stated" — no equivalent screen described



Full Arabic, right-to-left

Every screen, including this HOD dashboard, mirrors into a complete RTL layout — not just translated labels.

EXTEDO: "not mentioned in the material reviewed"

8 Strengths and weaknesses

PPR · strengths

- Covers the complete case, from application to certificate, in one system
- Validates against the official ICH DTD and the GCC Module 1 v1.5 criteria, not a reference copy
- Review by specialists per CTD module, with a lead reviewer and automatic assignment by workload
- Arabic and English, built for the department's own process
- AI assistant that answers from the dossier and cites page references
- Full validation documentation pack drafted, every requirement traced to an automated test
- Background validation queue, department-wide search, printable PDF reports, staff SSO and a read-only API added since the first comparison
- Owned by the department: no licence fees, and it can be changed quickly
- Web-only, so there is no client software to install

PPR · weaknesses

- Validation criteria are implemented but not yet benchmarked against a live reference validator — no EURSvalidator licence to test against
- The GCC regional DTD is still transcribed from the published specification, not SFDA's own distributed file
- Qualification protocol (IQ/OQ/PQ) and risk assessment are drafted but not yet executed or signed off
- No true in-PDF annotation — review notes are linked to a page, not drawn on the document itself
- Still a single web application; a background queue was added, but there is no distributed/multi-instance deployment
- No eCTD 4.0 or NeeS support (eCTD 4.0 is now detected and refused cleanly, not silently mishandled)
- Relies on a small team for support

EXTEDO · strengths

- Validation sets aligned with authorities, covering the GCC and kept current by the vendor, across every region it supports
- Proven at EMA and many national authorities
- Rich review tools: true annotation, highlighting and hyperlinking of PDFs; full-text search; lifecycle navigation
- Enterprise features: API, Active Directory, document management integration, 400+ users
- Vendor support, training and consulting (Agency Services)

EXTEDO · weaknesses

- No applicant portal, case workflow, decision or certificate in EURS itself
- Licence and subscription costs (not public), plus local partner fees
- Mainly a Windows desktop client; full web access only through EURSnext
- Arabic interface not mentioned
- Customisation depends on the vendor
- No public screenshots of the live product on EXTEDO's current marketing site (section 7)

9 Progress since v0.1 and what PPR still needs for production

The first comparison listed twelve gaps between PPR and the level of the EXTEDO tools for the parts the department needs. The table below carries the same numbering and tracks what has happened to each one since 18 October 2026.

#	Gap (v0.1)	Status	What happened / what is left	Priority
1	Reference DTD instead of the official one	Closed	The official ICH eCTD v3.2.2 DTD is bundled and checksum-verified (`apps/validation/dtd/README.md` , `test_bundled_dtd_is_the_official_ich_dtd`).	High
2	GCC Module 1 checked only for well-formedness	Closed	GCC M1 v1.5 implemented as rules GCC-01 to GCC-09 (`apps/validation/regions/gcc.py`). The regional DTD file itself is still a transcription — tracked below as a residual note, not a reopened gap.	High
3	No benchmark against a reference validator	Open	Procedure, sample-package generator and comparison tooling are built and tested (`docs/validation_benchmark.md` , `apps/validation/benchmark.py`). Not run: the department has no EURSvalidator evaluation licence. This is PQ-02 in the qualification protocol.	High
4	Synchronous validation on PythonAnywhere	Closed	`VALIDATION_MODE=queue` plus `run_validation_worker` as an always-on task; stale jobs are requeued and failures notify the company (`test_validation_queue.py`). Upload size limits for very large dossiers still need confirming in production (PQ-01).	High
5	No formal system validation	In progress	URS (23 requirements), risk assessment (12 hazards), an IQ/OQ/PQ protocol and an automated traceability matrix are drafted, with every requirement linked to tests. All three documents are "version 0.1, draft" with blank signature blocks and an empty execution log — qualification execution and HOD/QA/Ministry IT sign-off are the work still to do.	High
6	No cumulative lifecycle view	Closed	"Current dossier" tab applies replace/append/delete across all accepted sequences (`test_lifecycle_view.py`).	Medium
7	Validation report only on screen and as CSV	Closed	Printable short and long PDF validation reports added (`test_validation_report.py`).	Medium
8	No review annotations	Partly closed	Page-linked notes, optionally quoting selected text, collected and inserted into the module assessment (`test_document_notes.py`). This is notes alongside the page, not true in-PDF highlighting/markup — the original gap's "kept separate from the file" goal is met; the annotation-on-the-document itself is not.	Medium
9	Assistant tested only in offline mode	Open	A data-protection decision framework is written (`docs/assistant_data_protection.md`) and a smoke-test command exists, but the HOD has not yet recorded a decision on where the model runs, and no live-key test has been logged.	Medium
10	Search limited to one dossier	Closed	Full-text search across every dossier a reader may open, grouped by document, with a MySQL FULLTEXT index (`test_search.py`).	Low

11	No integration interface or single sign-on	**Closed**	Key-protected read-only API plus optional OpenID Connect SSO for staff (`docs/sso.md` , `test_api_and_sso.py`).	**Low**
12	No eCTD 4.0 or NeeS support	**Open**	An eCTD 4.0 package is now recognised and refused with a clear message instead of being mishandled. No actual eCTD 4.0 or NeeS validation support was added; still low priority while GCC authorities remain on eCTD 3.2.	**Low**

Summary: 7 of 12 closed outright, 1 partly closed, 1 in progress with documentation complete but unexecuted, 3 still open. The three open items (3, 9, 12) and the in-progress item (5) are the focus for the next cycle of work; of these, item 3 (the benchmark) and item 5 (qualification sign-off) are what stand between PPR and a defensible go-live decision.

10 Options and recommendation

Option	Description	Advantages	Disadvantages
A. PPR on its own	Continue PPR; run the benchmark and execute the qualification protocol (section 9, items 3 and 5).	Lowest cost; full control; Arabic; complete workflow; the design-level validation gap from v0.1 is already closed.	The team must still prove its GCC criteria against a reference validator and complete formal sign-off.
B. EXTEDO on its own	License EURS/EURSnext and EURSvalidator; manage the workflow outside the system (email, paper, spreadsheets).	Proven validation and review tools; vendor support.	No portal, assignment, decision or certificate; licence costs; English only.
C. Hybrid	PPR manages the case, portal and decisions; an EXTEDO validator or EURS is called through its API, or used alongside PPR for validation and review.	Best validation and review depth together with the department's own workflow.	Licence costs plus integration work; two systems to support.

Recommendation. Proceed with option A and keep option C open, same as v0.1 — but the next steps have narrowed:

1. Run the validation benchmark against EURSvalidator or another reference validator (gap 3 / PQ-02). This is now the single highest-value remaining step: the tooling is ready, only the reference run is missing. A trial or evaluation licence could be requested from EXTEDO or its regional partner for this purpose.
2. Execute the IQ/OQ/PQ qualification protocol in a production-like environment and obtain the HOD/QA/Ministry IT sign-off on the URS, risk assessment and qualification protocol (gap 5).
3. Record the HOD's decision on where the dossier assistant's language model runs, and log a live-key test before relying on written answers in production (gap 9).
4. If the benchmark shows differences that are costly to close, or dossier volumes grow beyond what PythonAnywhere can handle, move to option C and let PPR call an external validator while keeping its workflow, Arabic interface and assistant.
5. Ask EXTEDO to confirm the points marked "Not stated" in this report, and to provide a price for a GCC validation-set subscription or evaluation licence, so the options can be compared on cost. This has not yet been done.

A Sources

Vendor material is quoted as the vendor's own claims. EXTEDO links re-checked 21 October 2026; the PPR comparison report v0.1 was based on material retrieved 18 October 2026.

PPR

- PPR project repository: `README.md`, `apps/applications`, `apps/validation`, `apps/assistant`, `apps/accounts`, `tests/` (state as of 21 October 2026)
- PPR-CMP-001 v0.1, Comparative assessment report (18 October 2026) — the version this report updates
- `docs/validation/urs.md`, `risk_assessment.md`, `iq_oq_pq.md`, `traceability.md` — PPR-VAL-URS, PPR-VAL-RA, PPR-VAL-QP (version 0.1, draft)
- `docs/validation_benchmark.md` — benchmark procedure against a reference validator
- `docs/sso.md`, `docs/assistant_data_protection.md`

EXTEDO

- EXTEDO, EURS Product Information (flyer): extedo.com/hubfs/Documents/Flyers/EXTEDO_EURS_Product_Information.pdf
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- European Medicines Agency, eSubmission: EURS: esubmission.ema.europa.eu/tiges/eurs.htm
- Pharmaknowl, SFDA eCTD module requirements (GCC specification version): pharmaknowl.com/sfda-ectd-modules-requirements
- Biomapas, eCTD implementation across the MENA region: biomapas.com/ectd-implementation-across-mena-region